

ICD-9-CM codes with no match in the target system

Of the 354 CCS-related ICD-9-CM codes, 9 have no ICD-10-CA match and 52 have no ICDA-8 match in the reference crosswalk. For these codes the correct output is no mapping, and any mapping the algorithm produces for them is a false positive.

The algorithm uses two numbers. One is how often a pair of codes appears on the same patient record, the other is the cosine similarity between the two code labels, taken here from ClinicalBERT. Both are described below for the two groups together and then for each code individually, with the codes that do have a match included for comparison.

Algorithm output

On ICD-9-CM to ICD-10-CA the algorithm produces a mapping for all 9 codes that should receive none, so all 9 are false positives and none are correctly rejected. On ICD-9-CM to ICDA-8 it returns no mapping for all 52, and it also returns no mapping for 39 of the 302 codes that do have a valid match.

Group summary

Co-occurrence. A code appears in the co-occurrence file only if it shares a record with at least one target code, so a code absent from the file has no data rather than a low count. Partners is the number of different target codes a code appears with, and the last five columns describe each group by the highest count reached by any one of its codes. Pairs below a frequency of 6 were removed when the co-occurrence data was built, and the lowest count present is 7.

Target system	Group	Codes	Not in the file	Median partners	Highest count, lowest	25th	Median	75th	Highest count, highest
ICD-10-CA	no match	9	0	161	16	131	1,783	4,862	13,431
ICD-10-CA	has a match	345	7	209	7	264	1,288.5	5,633.5	238,548
ICDA-8	no match	52	52	no data	no data	no data	no data	no data	no data
ICDA-8	has a match	302	32	35.5	7	102.8	325.5	827	19,099

For ICD-10-CA the middle value of the highest count is 1,783 in the group with no match and 1,288.5 in the group that has one. For ICDA-8 none of the 52 codes with no match appear in the file.

Cosine similarity. Each ICD-9-CM code is scored against every target code, 2038 for ICD-10-CA and 858 for ICDA-8, so the average column covers all of those scores across all codes in the group. The last five columns describe each group by each code's single highest score, which is the score that determines whether a mapping is made.

Target system	Group	Codes	Average of every score	Highest score, lowest	25th	Median	75th	Highest score, highest
ICD-10-CA	no match	9	0.807	0.853	0.915	0.915	0.942	0.955
ICD-10-CA	has a match	345	0.820	0.861	0.914	0.930	0.948	0.981
ICDA-8	no match	52	0.830	0.883	0.931	0.949	0.964	0.985
ICDA-8	has a match	302	0.828	0.865	0.950	0.975	1.000	1.000

On both crosswalks the group with no match sits inside the range of the group that has one.

ICD-9-CM to ICD-10-CA

Co-occurrence for each of the 9 codes.

Code	Label	Partners	Min	25th	Median	75th	Max	Most frequent partner
338	pain not elsewhere classified	827	7	18	49	181.5	13,431	E11, Type 2 diabetes mellitus
515	postinflammatory pulmonary fibrosis	306	7	13	24	53.8	5,194	J84, Other interstitial pulmonary diseases
239	neoplasms of unspecified nature	585	7	14	34	91	4,862	I10, Essential (primary) hypertension
327	organic sleep disorders	349	7	12	23	49	3,108	G47, Other sleep disorders
249	secondary diabetes mellitus	161	7	12	18	38	1,783	E11, Type 2 diabetes mellitus
209	neuroendocrine tumors	124	7	11.8	17	31.8	506	C78, Secondary malignant neoplasm of respiratory and digestive organs
175	malignant neoplasm of male breast	24	7	8	10.5	22.5	131	C79, Secondary malignant neoplasm of other and unspecified sites
339	other headache syndromes	58	7	9.2	15	17.8	54	E11, Type 2 diabetes mellitus
445	atheroembolism	2	7	9.2	11.5	13.8	16	I74, Arterial embolism and thrombosis

Code 338, pain not elsewhere classified, appears with 827 different ICD-10-CA codes, and with E11, type 2 diabetes mellitus, 13,431 times. Code 515 reaches 5,194 and code 239 reaches 4,862. 5 of the 9 exceed 1,288.5, the middle value of the highest count across the 345 codes that do have a match. Two are low: code 445, atheroembolism, has 2 partners and a highest count of 16, and code 339, other headache syndromes, has 58 partners and a highest count of 54.

The codes with the highest counts are ones recorded across a wide range of patients, such as pain, sleep disorders and neoplasms of unspecified nature.

Cosine similarity for each of the 9 codes. Each row summarises that code's 2038 scores.

Code	Label	Average	Min	25th	Median	75th	Max	Closest target code
175	malignant neoplasm of male breast	0.825	0.641	0.796	0.826	0.849	0.955	C30, Malignant neoplasm of nasal cavity and middle ear
249	secondary diabetes mellitus	0.817	0.663	0.774	0.817	0.859	0.951	O24, Diabetes mellitus in pregnancy
239	neoplasms of unspecified nature	0.831	0.662	0.804	0.833	0.859	0.942	D15, Benign neoplasm of other and unspecified intrathoracic organs
515	postinflammatory pulmonary fibrosis	0.852	0.694	0.823	0.858	0.886	0.932	N72, Inflammatory disease of cervix uteri
209	neuroendocrine tumors	0.836	0.693	0.808	0.842	0.867	0.915	A74, Other diseases caused by chlamydiae
327	organic sleep disorders	0.770	0.541	0.725	0.773	0.823	0.915	P78, Other perinatal digestive system disorders
339	other headache syndromes	0.816	0.648	0.786	0.820	0.851	0.915	A69, Other spirochaetal infections
338	pain not elsewhere classified	0.805	0.685	0.785	0.807	0.827	0.893	R52, Pain, not elsewhere classified
445	atheroembolism	0.711	0.552	0.686	0.713	0.736	0.853	M66, Spontaneous rupture of synovium and tendon

The highest score for these 9 runs from 0.853 to 0.955, against 0.861 to 0.981 for the 345 codes that do have a match. In 8 of the 9 the closest ICD-10-CA code describes a different condition. Code 175, malignant neoplasm of male breast, is closest to C30, malignant neoplasm of nasal cavity and middle ear, at 0.955, and code 515, postinflammatory pulmonary fibrosis, is closest to N72, inflammatory disease of cervix uteri. These pairs share wording rather than clinical meaning. The exception is code 338, whose closest target R52 carries the same label, while the crosswalk records no match for it.

ICD-9-CM to ICDA-8

None of the 52 codes appear in the co-occurrence file, so there is no co-occurrence data to describe for them. The algorithm requires co-occurrence data before accepting a pair, which is why it returns no mapping for all 52. The same gap covers 32 of the 302 ICD-9-CM codes that do have a valid ICDA-8 match, and accounts for the 39 mappings it does not produce.

Cosine similarity. The highest score for the 52 runs from 0.883 to 0.985, against 0.865 to 1.000 for the 302 codes that do have a match. The four highest and the four lowest are below.

Code	Label	Average	Min	25th	Median	75th	Max	Closest target code
164	malignant neoplasm of thymus heart and mediastinum	0.813	0.584	0.776	0.810	0.846	0.985	142, malignant neoplasm of salivary gland
175	malignant neoplasm of male breast	0.816	0.578	0.782	0.816	0.848	0.984	174, malignant neoplasm of breast
237	neoplasm of uncertain behavior of endocrine glands and nervous system	0.821	0.594	0.785	0.822	0.855	0.984	238, neoplasm of unspecified nature of eye brain and other parts of nervous system
179	malignant neoplasm of uterus part unspecified	0.812	0.633	0.780	0.810	0.842	0.980	149, malignant neoplasm of pharynx unspecified
588	disorders resulting from impaired renal function	0.820	0.700	0.801	0.822	0.840	0.901	515, pneumoconiosis due to silica and silicates
312	disturbance of conduct not elsewhere classified	0.775	0.678	0.751	0.775	0.797	0.895	306, special symptoms not elsewhere classified
625	pain and other symptoms associated with female genital organs	0.762	0.540	0.727	0.768	0.807	0.893	781, other symptoms referable to nervous system and special senses
315	specific delays in development	0.777	0.671	0.752	0.776	0.800	0.883	066, late effects of viral encephalitis

Threshold scenarios

A similarity cut-off high enough to exclude all 9 ICD-10-CA codes would have to sit above 0.955, which also excludes 286 of the 345 codes that are correct. On ICDA-8 the equivalent cut-off is 0.985, which excludes 175 of 302. A co-occurrence cut-off high enough to exclude the 9 would have to sit above 13,431, which excludes 293 of the 338 matched codes present in the file. Requiring both conditions does not change this, since the 9 ICD-10-CA codes pass both, and the 52 ICDA-8 codes are excluded by the absence of co-occurrence data rather than by a threshold.

Threshold implementation

Step 1 of the Methods defines the similarity cutoff as the maximum similarity score for a code multiplied by the threshold value, which makes it relative to the code being mapped. Table 2 of the same document describes the thresholds as the minimum cosine similarity required for a match. The code implements the Step 1 definition.

Because the cutoff is a fraction of each code's own highest score, the highest scoring target of every code clears it. The lowest maximum on this crosswalk is 0.853, which gives that code a cutoff of 0.844 at a threshold of 0.990 and 0.853 at 1.0, and its highest scoring target clears both.

Step 2 selects each code's top N partners ranked by co-occurrence frequency, so the count is not compared against a value. The lowest maximum count in the file is 7, and it is selected on the same terms as the code with 238,548.

Under these definitions a pair is not excluded for a low score on either signal. A code produces no mapping when the Step 3 chapter distance filter removes all of its candidates, or when mapping algorithm 3, the combination selected in the Methods, finds no co-occurrence data for it. The second applies to all 52 ICDA-8 codes.